MATH329 · OPERATIONS RESEARCH

Midterm 472 Practice

13/04/2026 · 60 minutes · 30 marks

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01 Questions

Answer each question in the space below it. The full solution is in a separate file.

Q1 TRUE / FALSE · 7 M

Answer true or false for all of the following:

1. Graphical solutions are only applicable to LP problems with more than three variables.
2. Operations Research was first initiated during World War I in England to optimize war material utilization.
3. In a linear programming problem, the objective function or the constraints must be linear.
4. The Simplex tableau is used to organize the objective function and constraint equations for the algorithm.
5. The solution of the LPP may be compatible with its constraints imposed by the available resources.
6. The Simplex Algorithm requires the coefficients of the starting basic variables to form an identity matrix.
7. In Simplex method, if no variable can enter the basis, the problem has no feasible solution.

Q2 GRAPHICAL · 8 M

Solve the following problem graphically:

Maximum $Z = 5x_1 + 3x_2$

S.to: $2x_1 + 3x_2 \leq 12$

$2x_1 - x_2 \leq 8$

$-2x_2 \geq -4$

$x_1 \geq 0, x_2 \geq 0$

Q3 STANDARD FORM + SIMPLEX · 5 + 10 M

Given the following problem:

Maximize $z = 16x_1 + 15x_2$

Subject to: $40x_1 + 31x_2 \leq 124$

$x_1 - x_2 \geq -1$

$x_1 \leq 3$

$x_1, x_2 \geq 0$

a) Write the problem in standard form.

b) Solve the problem using Simplex Algorithm (Stop at first Iteration "2 tables only").

Basic	x_1	x_2	s_1	s_2	s_3	Sol.
Z						
Z						